

Aditya Shirsatrao

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Solapur, Maharashtra, India

Professional Summary

B.Tech AI & Data Science student (CGPA 8.5/10) with hands-on experience building production-grade ML systems, NLP pipelines, and full-stack AI products across cybersecurity, observability, and legal-tech domains. Three-time national hackathon winner, IEEE first-author publication at ICCTWC 2026, and GDG AI/ML Lead mentoring 200+ students. Proficient in PyTorch, TensorFlow, HuggingFace, and RAG/LLM systems.

Education

N. K. Orchid College of Engineering & Technology	Solapur, Maharashtra
B.Tech in Artificial Intelligence and Data Science – CGPA: 8.5/10	Aug. 2023 – Present (Expected 2027)

Technical Skills

Machine Learning: PyTorch, TensorFlow 2.x, HuggingFace Transformers, Scikit-learn, OpenCV, NLP, Computer Vision, Transfer Learning, Model Optimization (Quantization, Distillation), RAG, LangChain, AI Agents, FAISS, Vector Embeddings
Programming Languages: Python (expert), Java, C++, JavaScript, SQL
Backend & Infrastructure: FastAPI, Node.js, PostgreSQL, Redis, Celery, Docker, Kubernetes, Kafka, Microservices
Core Competencies: DSA (366+ LeetCode), System Design, ML Systems Architecture, Anomaly Detection, Feature Engineering, Model Evaluation, Real-time Data Processing

Experience

Fullstack Web Development Intern	Unified Mentor Pvt. Ltd
Production ML & Distributed Systems Development	Jul. 2025 – Jan. 2026
– Architected an AI-powered code review platform with asynchronous GitHub PR processing; integrated Kafka queues and Redis caching, reducing analysis latency by 45% via parallel distributed workers and vectorized AST-based code search.	
– Built a distributed ML inference platform with autoscaling endpoints, model versioning, and Prometheus monitoring; improved throughput by 60% under concurrent workloads via request batching and inference caching.	
AI/ML Lead	Google Developer Groups on Campus (GDG-NKOCET)
Community Leadership & Mentorship	Sep. 2025 – Present
– Lead AI/ML initiatives for a 200+ student community, organizing workshops on production ML, LLMs, and MLOps best practices.	

ML Engineering Projects

SentinelX – AI Cybersecurity Platform | Python, FastAPI, DistilBERT, Whisper, Celery, Redis, PostgreSQL | CodeMay 2026

- Built a production-grade multi-channel threat detection platform analyzing phishing, scams, and social engineering across email, SMS, and voice using a multi-model pipeline: HuggingFace DistilBERT for NLP classification, OpenAI Whisper for speech-to-text, and Qwen 2.5-7B for LLM enrichment.
- Engineered a weighted 4-component risk scoring engine (NLP, behavioral, URL, and sender reputation signals) into a unified threat score, decoupled via async Celery + Redis task queues; won 1st Place at Hack to Future 3.0 (600+ teams, INR 50,000).

RAG Knowledge Assistant | FastAPI, Chroma, sentence-transformers, BM25, LangChain | CodeJun. 2026

- Built an LLM Q&A system over 500+ documents using hybrid retrieval (dense vector + BM25) with source citations, achieving 92% retrieval hit-rate on a 50-question eval set; runs offline with zero model downloads.

Production ML System: Network Intrusion Detection | TensorFlow, scikit-learn, ML Systems Design | CodeMar. 2026 – Apr. 2026

- Architected a complete ML system for real-time anomaly detection and multi-class threat classification on streaming network traffic.
- Engineered a feature pipeline extracting 40+ features, achieving 96% accuracy with <100ms inference latency on NSL-KDD and UNSW-NB15 (24+ attack types), 95%+ recall on novel attacks.
- Handled class imbalance with SMOTE; compared Random Forest, XGBoost, and LSTM for the precision-recall trade-off.

Multilingual Sentiment Analysis with BERT Fine-Tuning | PyTorch, HuggingFace Transformers, NLP | CodeApr. 2026

- Fine-tuned BERT-base-multilingual-uncased for 104+ languages achieving 92%+ accuracy on multilingual sentiment classification; published the model to the HuggingFace Model Hub.
- Applied 8-bit quantization, knowledge distillation, and batch processing, reducing inference latency by 35% while maintaining accuracy.

Achievements & Research

IEEE First-Author Publication on multilingual sentiment analysis and cross-lingual classification, presented at ICCTWC 2026.
Hackathon Champion: 1st Place Hack to Future 3.0 (600+ teams); 1st Runner-Up Orchathon 2K26 (800+ teams).
Open Source: 50+ PRs merged across Google (OSS-Fuzz), Microsoft, Apache (DataFusion), and Hedera (Hiero SDK).
Certifications: Kaggle AI Agents; JPMorgan Chase & Electronic Arts SWE Virtual Experience; NPTEL Java (Elite+Gold, 91/100).